

Aman Nindra

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EDUCATION

University of California, San Diego

La Jolla, CA

Bachelor of Science, Data Science, spec AI/ML

June 2028

Relevant Coursework: Data Structures, Advanced Programming, Computer Organization and Assembly Language, Algorithm Design and Analysis, Algebra, Calculus I–III, Probability & Statistics, **Unsupervised Machine Learning (Grad)**, Applied Data Science, Introduction to Research Methods

Awards: 2nd place, Undergraduate Research Opportunities Center Research (50+ projects)

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript/TypeScript, SQL, Bash/Shell

AI/ML: PyTorch, TensorRT, ONNX Runtime, Computer Vision, Natural Language Processing, Transformers, YOLO, NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV

GPU: NVIDIA Jetson, JetPack, ROS 2, Real-Time Inference, GPU Performance Profiling, Distributed Training

Cloud & HPC: AWS, Google Cloud, Linux, Docker, Kubernetes, Slurm, Terraform

Frameworks & Tools: React.js, Next.js, Flask, FastAPI, Three.js, Git, Firebase, MCP, Claude Code, Codex, Copilot

EXPERIENCE

Data Science Intern

July 2026 – August 2026

Lawrence Livermore National Laboratory

Research Poster

- Built the core of a multi-agent AI inspection system using image processing to align 3D X-ray CT scans of 3D-printed metal parts with their original designs and automatically measure structural rods.
- Automatically classified all **18,468** structural rods against the original design, identifying **118** missing and **256** broken candidates with each classification traceable to the source CT data.
- Developed an interactive 3D inspection tool (Streamlit, Three.js) on an open-source image-processing toolkit, enabling researchers to select individual rods in a color-coded model and trace defect labels to supporting internal images.

Undergraduate Researcher

May 2026 – Present

R & D Lab

Research Poster

- Built a real-time autonomous-vehicle perception pipeline using LaneATT with a ResNet-34 backbone, improving CULane lane-detection F1 from **0.77** to **0.79** through custom data augmentation.
- Trained a YOLOv11n object detector for pedestrians, vehicles, traffic lights, and stop signs by adapting the COCO dataset, achieving **0.650 mAP50** and **0.465 mAP50-95**.
- Profiled and optimized deep-learning inference on an NVIDIA Jetson Orin Nano Super, benchmarking ONNX Runtime with CUDA against TensorRT FP16 and reducing LaneATT latency by **4.55x** (**125.7 ms** to **27.6 ms**) and YOLO latency by **1.83x** (**30.3 ms** to **16.6 ms**).
- Integrated LaneATT and YOLOv11 inference into ROS 2 nodes, processing live stereo-camera feeds and publishing lane and object detections for real-time autonomous bicycle control.

Software Engineering Intern

May 2026 – August 2026

Valley Institute for Sustainability, Technology & Agriculture

Remote

- Architected a GPU-accelerated machine-learning platform on a **3-node Kubernetes cluster** with **70 TB** of persistent storage and **2 GPU worker nodes**, enabling **40+ professors** and **50+ PhD researchers** to train computer vision and NLP models.
- Built Terraform-based development environments that allowed researchers to configure CPU, RAM, and NVIDIA GPU allocations and provision reproducible PyTorch, TensorFlow, and Conda workloads through browser-based or remote VS Code environments.

PROJECTS

Multi-Label Emotion Classification with BERT | *PyTorch, Transformers, Scikit-learn* | [GitHub](#)

Jan. 2026

- Fine-tuned BERT for multi-label emotion classification across **28 categories** on **206K Reddit comments**, achieving **0.712 Macro F1** and **97.0% label-wise accuracy**.
- Developed per-class threshold calibration using grid search across **0.01–0.99** for each output label, replacing a global classification threshold and improving predictions for minority emotion classes.
- Built an evaluation pipeline measuring Macro F1, Jaccard similarity, label-wise accuracy, and exact-set accuracy, achieving **0.688 Jaccard similarity** and correctly predicting the complete emotion set for **47.7%** of samples.

Wildlife Camera-Trap Image Classification | *PyTorch, Vision Transformers, AWS* | [GitHub](#)

Nov. 2025 – Feb. 2026

- Placed **4th** on DrivenData's wildlife camera-trap leaderboard, reducing competition log loss from **2.18** to **0.67** by fine-tuning a **300M-parameter EVA-Large Vision Transformer** to classify **8 African species** across **16,488 images**.
- Trained EVA-Large using PyTorch DistributedDataParallel (DDP) and mixed-precision training across 4 NVIDIA Tesla V100 GPU's on AWS SageMaker, achieving **93.3% validation accuracy**.
- Built a per-class evaluation pipeline measuring precision, recall, F1, and top-3 accuracy each epoch, identifying blank-frame misclassification as the dominant error source and the only class below **0.80 F1** across four evaluated architectures.